

# Timestamp & Validation-Based Protocols DBMS

> **Definition: Timestamp Ordering Protocols** ensure conflict serializability without using locks by assigning a unique monotonic timestamp **TS(T<sub>i</sub>)** to each transaction upon start and enforcing operation ordering according to timestamp sequence.

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## 1. Detailed Technical Explanation

### 1. System Timestamps

Each data item **X** maintains two timestamps:

### 2. Basic Timestamp Ordering Rules

#### Read Operation `Read(X)` by Transaction **T<sub>i</sub>**:

#### Write Operation `Write(X)` by Transaction **T<sub>i</sub>**:

## 2. Validation-Based (Optimistic) Protocol

Used when read operations dominate write operations (low conflict rate). Execution is split into 3 phases:

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### Validation Test Condition:

For all **T<sub>k</sub>** committed before **T<sub>i</sub>**:

## 3. Core Concepts & Memory Keywords

- **Monotonic Timestamps:** Unique growing values (System clock or global counter).

#### 4. Must-Write Points for Exams

- Timestamp ordering protocols guarantee **freedom from deadlocks** because transactions are never made to wait indefinitely.

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#### 5. Quick Recall Flow

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